

Mortgage Risk Modeling

07. Project Report

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Project Scope	Binary classification for mortgage default risk and mortgage prepayment risk

1. Executive Summary

This project developed two predictive models to analyze key mortgage outcomes:

- **Default**, defined as loans that became seriously delinquent.
- **Prepayment**, defined as loans that were voluntarily paid off before maturity.

Using historical mortgage performance data, the project evaluated whether borrower, loan, and property characteristics could help predict these outcomes. Both are important to mortgage lenders and investors: default affects credit risk and servicing performance, while prepayment affects cash flow timing and portfolio runoff. Because these outcomes reflect different borrower behaviors and different dimensions of mortgage risk, they were modeled separately.

2. Data and Target Summary

Source data size	Summary
Geographic scope	Five states
Entries	2,071,611 rows
Unique loans	104,924 unique LOAN_ID values
Starting columns	108 columns
Default target rate	Approximately 2.35%
Prepayment target rate	Approximately 50.9%
Modeling implication	Default is a low-base-rate event and therefore harder to predict reliably than prepayment

3. Feature Engineering Results

Target design	Created separate binary targets for default and prepayment
Categorical simplification	Consolidated selected rare or overly granular categories to improve model usability
Mortgage insurance feature	Converted MI information into a simplified flag structure
Loan term feature	Grouped term values into practical bins such as 15-year, 30-year, and other
Derived ratios	Created additional variables such as DTI_LTV and DTI_UPB to capture interaction effects
Value transformation	Added value-based engineered fields including home-value style measures and log-style variants where appropriate

4. Default Model Results

Target	Default
Model name	Gradient Boosting
Threshold used	0.5
AUC-ROC	0.759
Recall	0.702
F1 score	0.098

Interpretation:

For the default model, recall is the most important metric because default is a rare event and the main goal is to identify as many actual default cases as possible. A recall of 0.702 means the model captures about 70.2% of true defaults, which is a solid result for a highly imbalanced problem.

The AUC-ROC of 0.759 supports that the model has good overall ability to separate higher-risk loans from lower-risk loans. The F1 score of 0.098 shows that precision is still limited, which is common in rare-event prediction. Overall, the model is more useful as a risk-screening tool than as a highly precise decision tool.

5. Prepayment Model Results

Target	Prepayment
Model name	Gradient Boosting
Threshold used	0.4
AUC-ROC	0.703
Accuracy	0.627
F1 score	0.696

Interpretation:

For the prepayment model, F1 score is an important metric because it reflects how well the model balances identifying prepaid loans while limiting incorrect positive predictions. An F1 score of 0.696 suggests a reasonably strong balance between precision and recall.

The AUC-ROC of 0.703 indicates that the model has good ability to distinguish between loans that are more likely and less likely to prepay. The accuracy of 0.627 shows that the model is correct 62.7% of the time overall, which provides useful information because prepayment is not an extremely imbalanced target. However, F1 score gives more insight into positive-class prediction quality, which is often more important when evaluating how well the model identifies prepaid loans. Overall, the model shows useful predictive performance for understanding prepayment behavior.

6. Limitations and Next Steps

Limitations

- The default target was highly imbalanced at about **2.35%**, which made default prediction more difficult and increased the risk of false positives.
- The models used only the available loan, borrower, and property variables, so important external factors such as interest rates and economic conditions were not included.
- Default and prepayment were modeled as binary outcomes, which simplifies the fact that mortgage performance changes over time.

Future Steps

- Test additional models such as **XGBoost, LightGBM, or CatBoost**.
- Add macroeconomic variables to better capture market conditions.
- Optimize the classification threshold based on business needs.
- Explore time-based modeling to better reflect when default or prepayment happens.

7. Conclusion

This project established a structured mortgage risk modeling workflow that treats default and prepayment as separate analytical problems. That design is appropriate because the two outcomes reflect different borrower behaviors and different portfolio implications. The project also produced reusable modeling artifacts that can support reporting, review, and future refinement.